

Muhammad Asim

Full-Stack Engineer • Backend • AI-Powered Products

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PROFESSIONAL SUMMARY

Final-semester Software Engineering student at Air University, Islamabad (CGPA: 3.61/4.0), with three years of hands-on experience building backend systems, full-stack web applications, and AI-powered products. Co-founder and software engineer at **Illume Estate**—an incubated, funded startup that won the **Tabeer.ai competition**, secured external funding, and was recognised on the Air University Department Wall of Fame. Designed and built **PakSentiment** — a production-grade, multi-server sentiment analysis platform with four independent services (NestJS, FastAPI, Go/Gin, Next.js) and a custom Python scraping library.

EDUCATION

BSc Software Engineering 2022 – 2026 (Expected)
Air University, Islamabad • Department of Creative Technologies • CGPA: 3.61 / 4.0

- Scholarship recipient — awarded for outstanding academic and entrepreneurial performance.
- Co-founded Illume Estate with classmates — ranked high-performing startup at AUBIC incubation centre.
- Team selected for Department Wall of Fame for entrepreneurial achievement.

Courses

CS50: Introduction to Computer Science • Harvard University (HarvardX)
Full-Stack Web Development & Python • Zero to Mastery Academy (ZTM)
AI Search Algorithm • Indus AI Week 2026

TECHNICAL SKILLS

Languages	Go, Python, TypeScript, JavaScript, C++, Lua
Backend	ExpressJs, NestJS, Node.js, FastAPI, Django, Gin (Go), REST APIs, Socket.io
Databases	PostgreSQL, MySQL, MongoDB, Redis, Redis Streams, SQL
ORMs	Prisma, Mongoose
Message Brokers	RabbitMQ
Payments	Stripe, PayPal
Frontend	Next.js, Angular, React, React Native
AI / ML	TensorFlow, scikit-learn, HuggingFace Transformers, NLP pipelines
Scraping	Scrapling (custom library), Colly (Go), Reddit API, Twitter API, Common Crawl
Architecture	SaaS multi-tenant, Microservices, Webhook patterns, Event-driven design
Cloud & Hosting	AWS EC2, Azure Web Apps, Azure Compute, Vercel, Netlify, Supabase, Firebase, Cloudinary
Agentic Tools	Cursor, Antigravity, Gemini CLI, GitHub Copilot
Tooling	Docker, Git, GitHub Actions, Neovim, Linux

PROFESSIONAL EXPERIENCE

Co-founder & Software Engineer

Dec 2024 – Feb 2026

Illume Estate • AUBIC Incubation Centre, Islamabad

- Co-founded Illume Estate with classmates — a web and AI services company registered and incubated at Air University's Business Incubation Centre (AUBIC).
- Ranked as a high-performing startup at AUBIC, secured external funding through competitive evaluation, and won the Tabeer.ai startup competition.
- Designed and developed client-facing web products and AI-powered solutions end-to-end, managing technical delivery and client relationships.
- Team recognised by the Air University Department of Computing & Technology — selected for the department Wall of Fame.
- Participated in national-level startup championships representing Air University.

Frontend Developer Intern

Jun 2024 – Sep 2024

Cognitive Health International • Islamabad, Pakistan (Remote)

- Led the full architectural migration of a production Angular application to Next.js, taking sole ownership of project structure, component design, and code conventions.
- Worked directly with the lead developer to define scalable frontend architecture patterns adopted across the entire codebase.
- Delivered a maintainable, production-ready Next.js application independently within a three-month engagement.

React Native Developer

Oct 2025 – Nov 2025

Meelio.ai • Remote (Contract)

- Contracted to develop mobile UI screens and reusable components for Meelio.ai's React Native application.
- Developed OCR (Optical Character Recognition) for receipts and bills.

PROJECTS

PakSentiment — Final Year Project [\[NestJS\]](#) [\[FastAPI\]](#) [\[Go/Gin\]](#) [\[Next.js\]](#) [\[Redis\]](#) [\[NLP\]](#)

- Architected a multi-server sentiment analysis platform with four independent services: NestJS backend for auth and user flows, FastAPI AI services server for NLP inference, Go/Gin server for concurrent web scraping using Colly, and a Next.js frontend.
- Designed async inter-service communication using webhook patterns and Redis Streams — Go server acknowledges tasks with a job ID, processes asynchronously, then calls back to NestJS on completion.
- Authored a custom Python scraping library (Scrapling) to crawl Reddit, Twitter, and the open web, feeding structured data into the AI summarisation pipeline.
- Integrated HuggingFace Transformer models for sentiment classification and result summarisation, returning AI-generated insights per query.

pixl-go — Desktop Drawing Application [\[Go\]](#) [\[Fyne\]](#) [\[Cross-platform\]](#)

- Built a Windows Paint-style desktop drawing application entirely in Go using the Fyne UI framework — cross-platform, lightweight, zero external dependencies.
- Demonstrates proficiency in Go beyond web services — including GUI event handling, canvas rendering, and native desktop packaging.

Wails — Open Source Contribution [\[Go\]](#) [\[Open Source\]](#)

- Contributed to the Wails open source project — a Go framework for building cross-platform desktop apps with web frontends.
- Improved developer-facing documentation to help engineers integrate Go backends with modern web frameworks.

Ludo Game — C++ [\[C++\]](#) [\[graphics.h\]](#) [\[WinBGI\]](#)

- Developed a fully playable Ludo game with a complete graphical UI using the graphics.h and WinBGI graphics library.
- All animations, graphics, and visual effects are entirely code-generated using mathematics — no external assets or sprite files.
- Sole developer responsible for game logic, rendering pipeline, turn management, and player interaction from scratch.

ACHIEVEMENTS

- Won the **Tabeer.ai Startup Competition** — national-level startup championship representing Air University.
- Awarded Laptop in Prime Minister's Laptop Scheme (High Academic Performance)
- **Hackathon Winner** — 4th semester university-level hackathon.
- **Scholarship Recipient** — awarded by Air University for outstanding academic and entrepreneurial performance.
- **Department Wall of Fame** — team recognised by Air University CT Department for entrepreneurial achievement.
- Completed freelance web and AI development projects for external clients alongside full-time studies.